

# Lab Assignment 03



Inspiring Excellence

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|-------------------------|-------------------------|
| <b>Course Code:</b>     | CSE111                  |
| <b>Course Title:</b>    | Programming Language II |
| <b>Topic:</b>           | String, Array           |
| <b>Number of Tasks:</b> | 12                      |

## String

**Task - 01:** Write a Java program that will take one string input from the user. Then check and print whether it is a palindrome.

| Sample Input | Output           |
|--------------|------------------|
| Java         | Not a palindrome |
| madam        | Palindrome       |

**Task - 02:** Write a Java program that takes a string input in small letters from the user and prints the previous alphabet in sequence for each alphabet found in the input.

| Sample Input | Output |
|--------------|--------|
| wxyz         | vwxy   |
| thecow       | sgdbnv |
| abcd         | zabc   |

**Task - 03:** Write a Java program that will ask the user to input a string (containing exactly one word). Then your job is to print subsequent substring of the input string.

| Sample Input | Output                                    |
|--------------|-------------------------------------------|
| BANGLA       | B<br>BA<br>BAN<br>BANG<br>BANGL<br>BANGLA |
| DREAM        | D<br>DR<br>DRE<br>DREA<br>DREAM           |

**Task - 04:** Write a Java program that will ask the user to input a word in small letters where each of its alphabets is unique and has not been entered before by the user. If the user does input a word that consists of duplicate alphabets, the program should reject the user's input and ask for another word.

| Sample Input            | Output                                                                                                                                                                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fahim                   | You entered fahim.                                                                                                                                                                                                                             |
| farah<br>akbor          | "a" has been counted 2 times in the word "farah".<br>Please enter another word.<br>You entered akbor.                                                                                                                                          |
| alanna<br>ronan<br>john | "a" has been counted 3 times in the word "alanna".<br>"n" has been counted 2 times in the word "alanna".<br>Please enter another word.<br>"n" has been counted 2 times in the word "ronan".<br>Please enter another word.<br>You entered john. |

**Task - 05:** Write a Java program that takes TWO string inputs (containing exactly one word in each string) from the user. Concatenate those two strings with a single space in between them. Generate a number which is the sum of all the letters in that concatenated string where A = 65, Z = 90, a = 97, and z = 122. Your task is to print that concatenated string and the number generated from that string.

| Sample Input           | Output                      |
|------------------------|-----------------------------|
| Hello123<br>Wo%%rld    | Hello123 Wo%%rld<br>1020    |
| Ja12-va<br>CHOWD+ HURY | Ja12-va CHOWD+ HURY<br>1087 |

**Task - 06:** Given a string, create and print a new string with all the consecutive duplicates removed.

| Sample Input    | Output   |
|-----------------|----------|
| ABBCCCCBBAB     | ABCBAB   |
| AAABBBBCDDBBECE | ABCDBECE |

## Array

**Task - 01:** Write a Java program that will take an integer number N from the user and create an integer array by taking N numbers from the user. Then take another number from the user and create a new array by removing that number from the input array. Finally, print the new array.

| Sample Input                                                 | Sample Output                                                |
|--------------------------------------------------------------|--------------------------------------------------------------|
| N = 5<br>23<br>100<br>0<br>56<br>-34<br>Remove Element = 100 | Input array:<br>23 100 0 56 -34<br>New array:<br>23 0 56 -34 |
| N = 4<br>-5<br>10<br>2<br>-7<br>Remove Element = 43          | Input array:<br>-5 10 2 -7<br>Element not found              |

**Task - 02:** Write a program that reads 5 numbers into an array and prints the smallest and largest number and their location in the array.

| Sample Input            | Sample Output                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------|
| 7<br>13<br>2<br>10<br>6 | The largest number 13 was found at location 1.<br>The smallest number 2 was found at location 2.  |
| 2<br>4<br>-5<br>12<br>3 | The largest number 12 was found at location 3.<br>The smallest number -5 was found at location 2. |

**Task - 03:** Write a Java program that asks the user for the length of an array and then creates an integer array of that length by taking inputs from the user. Then, reverse the **original array without** creating any new array and print it. [**In-place reverse**]

| Sample Input                                                     | Sample Output    |
|------------------------------------------------------------------|------------------|
| Enter the length of the array: 5<br>7<br>-31<br>344<br>97<br>100 | 100 97 344 -31 7 |

**Task - 04:** Write a Java program that will take an integer number N from the user and create an integer array by taking N numbers from the user. Print how many times each number appears in the array.

| Sample Input                              | Sample Output                                                |
|-------------------------------------------|--------------------------------------------------------------|
| N = 5<br>6<br>15<br>14<br>15<br>6         | 6 - 2 times<br>15 - 2 times<br>14 - 1 times                  |
| N = 6<br>-5<br>10<br>14<br>10<br>-7<br>10 | -5 - 1 times<br>10 - 3 times<br>14 - 1 times<br>-7 - 1 times |

**Task - 05:** Write a Java program that asks the user the length of an array (N) then takes N number of integers as elements for the array as input. First, remove the consecutive duplicate elements from the original array to form a new array. Then print the number of elements removed from the original array.

| Sample Input                                                                             | Sample Output                                 |
|------------------------------------------------------------------------------------------|-----------------------------------------------|
| N = 8<br>Please enter the elements of the array:<br>5<br>2<br>1<br>1<br>2<br>3<br>3<br>3 | New Array : 5 2 1 2 3<br>Removed elements : 3 |

**Task - 06:** Write a program that asks the user how many numbers to take. Then takes that many numbers in an array and prints the median value.

[How to Find the Median Value: <http://www.mathsisfun.com/median.html>]

| Sample Input                    | Sample Output                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 5<br>10<br>50<br>40<br>20<br>30 | The median is 30.<br><br><b>Explanation:</b> 30 falls in middle 10, 20, 30, 40, 50                              |
| 4<br>30<br>10<br>40<br>20       | The median is 25.<br><br><b>Explanation:</b> $(20+30)/2=25$ (average of two middle values from 10, 20, 30, 40). |